Stress and Strain Energy Calculations

Problem

F1_1: A bedplate is secured by steel holding-down bolts with a diameter of 20 mm and a length of 180 mm. Each bolt is tightened onto a steel sleeve with an inside diameter of 25 mm, an outside diameter of 50 mm, and a length of 100 mm. After tightening, the tensile force in the bolt is 4 kN. Calculate the following:

A. The stress in each bolt
B. The strain energy stored in the bolt material
C. The stress in the sleeve
D. The change in length of the sleeve

Note: The modulus of elasticity for steel is 200 GPa.

Solution

A. Stress in Each Bolt

The stress in the bolt, σ_b, is calculated using the formula:

$$\sigma_b = \frac{F}{A}$$

where:

- $F = 4\,\text{kN} = 4000\,\text{N}$ (tensile force),
- $A = \pi\left(\frac{d}{2}\right)^2 = \pi\left(\frac{20\,\text{mm}}{2}\right)^2 = 314.16\,\text{mm}^2 = 3.1416 \times 10^{-4}\,\text{m}^2$ (cross-sectional area of the bolt).

Substituting the values:

$$\sigma_b = \frac{4000}{3.1416 \times 10^{-4}} = \boxed{12.73\,\mathrm{MPa}}\ 2.5\ marks$$

B. Strain Energy Stored in the Bolt Material

The strain energy, SE, is given by:

$$SE = \frac{\sigma_b^2}{2E} \cdot V$$

where:

- $E = 200\,\mathrm{GPa} = 200 \times 10^9\,\mathrm{Pa}$ (modulus of elasticity),
- $V = A \cdot L = 3.1416 \times 10^{-4} \cdot 0.18 = 5.6549 \times 10^{-5}\,\mathrm{m}^3$ (volume of the bolt).

Substituting the values:

$$SE = \frac{(12.73 \times 10^6)^2}{2 \cdot 200 \times 10^9} \cdot 5.6549 \times 10^{-5} = 0.0229\,\mathrm{J} = \boxed{22.9\,\mathrm{mJ}}\ 2.5\ marks$$

C. Stress in the Sleeve

The stress in the sleeve, σ_s, is calculated using the formula:

$$\sigma_s = \frac{F}{A_s}$$

where:

- $A_s = \pi\left(\frac{D_o^2 - D_i^2}{4}\right) = \pi\left(\frac{50^2 - 25^2}{4}\right) = 1472.62\,\mathrm{mm}^2 = 1.4726 \times 10^{-3}\,\mathrm{m}^2$ (cross-sectional area of the sleeve).

Substituting the values:

$$\sigma_s = \frac{4000}{1.4726 \times 10^{-3}} = \boxed{2.72\,\mathrm{MPa}}\ 2.5\ marks$$

D. Change in Length of the Sleeve

The change in length, ΔL_s, is given by:

$$\Delta L_s = \frac{\sigma_s \cdot L_s}{E}$$

where:

- $L_s = 100\,\text{mm} = 0.1\,\text{m}$ (length of the sleeve).

Substituting the values:

$$\Delta L_s = \frac{2.72 \times 10^6 \cdot 0.1}{200 \times 10^9} = 1.36 \times 10^{-6}\,\text{m} = \boxed{0.00136\,\text{mm}}\ 2.5\ \textit{marks}$$

Final Results

- **Stress in each bolt:** $12.73\,\text{MPa}$
- **Strain energy stored in the bolt material:** $0.0229\,\text{J}$
- **Stress in the sleeve:** $2.72\,\text{MPa}$
- **Change in length of the sleeve:** $0.00136\,\text{mm}$